otherwise there is little clear advantage and a potentially significant price to be paid, since HMC methods only work by rejecting steps. For reading on Hybrid Monte Carlo methods, the reader should consult [110, 188] and other references mentioned in Appendix C.

- *Kinetic Monte-Carlo (KMC) methods* are a broad family of multiscale sampling methods which use a local sampling mechanism to explore basins while invoking an Arrenhius law to estimate the rates or timescales for transitions between basins (we can think of basins as neighborhoods of potential energy minima). The Arrenhius law relates the rate constant k associated to a particular state-to-state transition to the exponential of the of the "activation energy" which is defined as the difference ΔE between the energy at a local minimum and that at the energy barrier between the states

$$k \sim Ae^{-\Delta E/\mathrm{k_B T}}$$

  In a KMC method, it is typically assumed that various possible state-to-state transitions from a given state are well modelled by the Arrenhius law and then molecular dynamics is used to calculate the prefactor A and energy difference ΔE in order to understand the timescales and relative probabilities of different rare events. A Markov state model can be developed to help understand the global dynamics and simplify the model as a whole. For references on many interesting approaches to this important topic, the reader is referred to [36, 42, 137, 149, 391].
- *Andersen Thermostat.* Of particular interest is the simple and useful *Andersen thermostat* [11]. This method works by selecting atoms at random and randomly perturbing their momenta in a way consistent with prescribed thermodynamic conditions. It has been rigorously proven to sample the canonical distribution [114].

8.4 Ergodic Stochastic-Dynamic Thermostats

In this section, we consider the combination of stochastic perturbation with a deterministic thermostat. Methods constructed in this way can be ergodic for the canonical distribution while also providing flexibility in way equilibrium is achieved. We distinguish in (6.16) between multiplicative noise, where $\boldsymbol{B} = \boldsymbol{B}(z)$ varies with z and additive noise, where $\boldsymbol{B}$ is constant. In our treatment of this topic we will only consider additive noise. The presence of multiplicative noise may complicate discretization. As we shall see, the reliance on additive noise improves the performance of discretization schemes.

To give this some specificity for molecular dynamics, we write the stochastic dynamical model as

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{8.22}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{\Phi}(\boldsymbol{\xi}, z) - \boldsymbol{\Gamma}\boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\mathrm{d}\mathbf{W}, \tag{8.23}$$

$$\mathrm{d}\boldsymbol{\xi} = \boldsymbol{g}(\boldsymbol{\xi}, \boldsymbol{q}, \boldsymbol{p})\mathrm{d}t - \hat{\boldsymbol{\Gamma}}\boldsymbol{\xi}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\hat{\boldsymbol{M}}^{1/2}\hat{\boldsymbol{\Gamma}}^{1/2}\mathrm{d}\hat{\mathbf{W}}. \tag{8.24}$$

Here $\boldsymbol{\Gamma}, \hat{\boldsymbol{\Gamma}}$ are symmetric, square matrices of appropriate dimension, $\mathbf{W}$ is a vector of $\mathrm{N_c}$ Wiener processes, and $\hat{\mathbf{W}}$ is a vector of m Wiener processes, all independent of each other. We assume that the deterministic thermostat represented by the $\boldsymbol{\Phi}$ and $\boldsymbol{g}$ terms preserves the extended distribution $\hat{\rho}(\boldsymbol{\xi})\rho_\beta$, where $\hat{\rho}(\boldsymbol{\xi}) = \exp\left(-\beta\boldsymbol{\xi}^T\hat{\boldsymbol{M}}^{-1}\boldsymbol{\xi}/2\right)$. The matrix $\hat{\boldsymbol{M}}$ is symmetric and positive definite; one could take it to be diagonal.

The additivity property of thermostats mentioned above holds also for stochastic thermostats, since the Fokker-Planck equation is constructed from linear operations on the vector fields involved, thus it is enough to require that the newly introduced terms satisfy appropriate fluctuation-dissipation relationships so that they are compatible with the extended Gibbs distribution of the deterministic part. That is, we require

$$\mathrm{d}\boldsymbol{p} = -\boldsymbol{\Gamma}\boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B}T}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\mathrm{d}\mathbf{W}$$

to preserve the distribution with density $\bar{\rho} = \exp\left(-\beta\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2\right)$, and

$$\mathrm{d}\boldsymbol{\xi} = -\hat{\boldsymbol{\Gamma}}\boldsymbol{\xi}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\hat{\boldsymbol{M}}^{1/2}\hat{\boldsymbol{\Gamma}}^{1/2}\mathrm{d}\hat{\mathbf{W}}$$

to preserve $\hat{\rho}$. These two problems are essentially the same. Considering the former, the stationary Fokker-Planck equation reads

$$\underbrace{\nabla_{\boldsymbol{p}} \cdot (\boldsymbol{\Gamma}\boldsymbol{p}\bar{\rho})}_{\text{drift}} + \underbrace{\beta^{-1}\mathrm{Tr}\left(\boldsymbol{\Gamma}^{1/2}\boldsymbol{M}^{1/2}\nabla_{\boldsymbol{p}}^2\bar{\rho}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\right)}_{\text{diffusion}} = 0.$$

The drift term here equates to

$$\mathrm{Tr}(\boldsymbol{\Gamma})\bar{\rho} + (-\beta)\boldsymbol{p}^T\boldsymbol{\Gamma}\boldsymbol{M}^{-1}\boldsymbol{p}\bar{\rho}.$$

For the diffusion term, we calculate

$$\nabla\bar{\rho} = -\beta\bar{\rho}\boldsymbol{M}^{-1}\boldsymbol{p},$$

thus

$$\nabla^2\bar{\rho} = \bar{\rho}[\beta^2\boldsymbol{M}^{-1}\boldsymbol{p}\boldsymbol{p}^T\boldsymbol{M}^{-1} - \beta\boldsymbol{M}^{-1}],$$